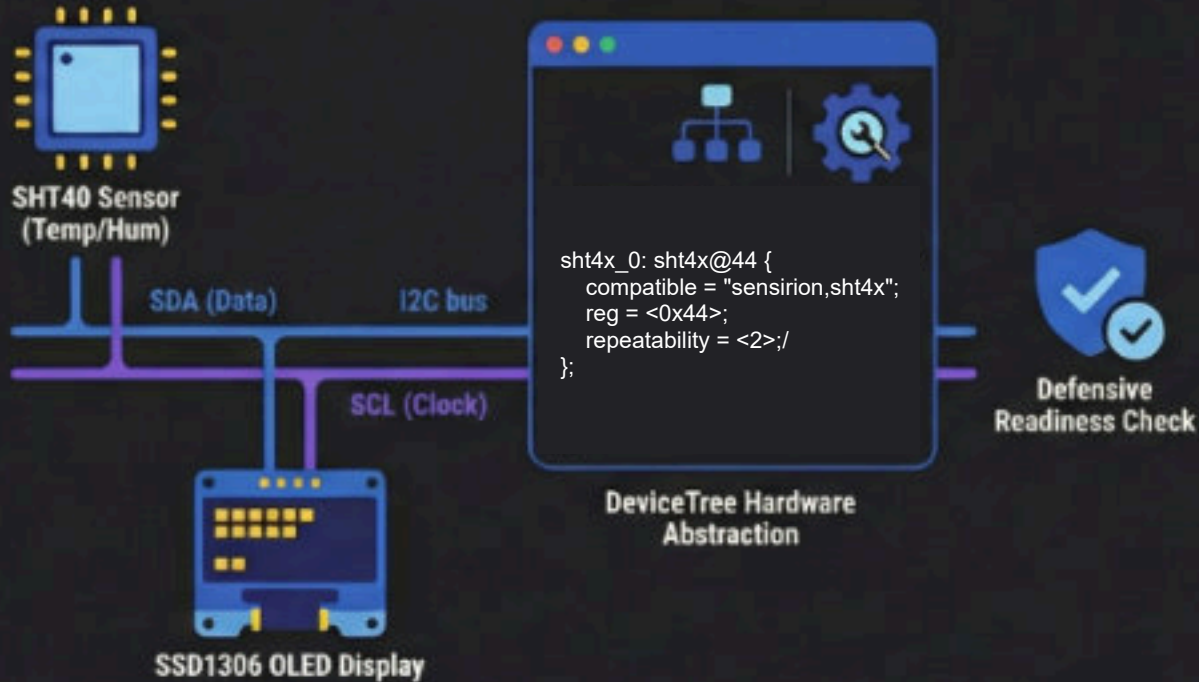


## Phase 1: Hardware Configuration & Binding



- I2C Bus Addressing & Sharing**

The SHT40 and OLED share SDA/SCL lines, uniquely identified by independent I2C addresses.

- DeviceTree Hardware Abstraction**

Hardware is defined in DeviceTree (node `sht4x_0`), allowing the application to remain portable.

- Defensive Readiness Check**

The system verifies device readiness before initialization to catch wiring or configuration errors.

## Phase 2: The Measurement & Display Loop



### Standardized Sensor Sampling:

`'sensor_sample_fetch()'` triggers the measurement, while `'sensor_channel_get()'` retrieves specific temperature and humidity data.



### Data Normalization

New values are converted from the "sensor Value" struct into doubles for human readable



- Synchronized OLED Update**

Measurements are formatted into strings and pushed to the character framebuffer every second.

### Key Zephyr API Functions



#### `sensor_sample_fetch()`

Triggers hardware measurement and retrieves raw bus data.



#### `sensor_channel_get()`

Extracts specific channels like temperature or humidity.



#### `sensor_value_to_double()`

Converts Zephyr's internal sensor format to standard floating point.